PostgreSQL version sensitivity (Tier 2, item 36)

The same 22 TPC-H queries on PostgreSQL 14, 15, 17, 18, over byte-identical data (moved by `pg_dump`, so only the engine differs). Tier 3 showed the plan is hardware-dependent; this asks whether it is version-dependent too — which matters because the entire EC2 dataset comes from a single version.

12 of 22 queries get a different plan on at least one version.

Runtime relative to PostgreSQL 15 (median over queries):

version	median runtime ratio	queries with a different plan vs 15
14	0.796x	4/22
15	1.000 (reference)	—
17	0.708x	2/22
18	0.740x	9/22

Per query

query	t14 (ms)	t15 (ms)	t17 (ms)	t18 (ms)	distinct plans
q1	1282	1813	1252	1173	1
q10	389	448	476	262	4
q11	140	174	92	475	2
q12	428	580	343	301	1
q13	1156	1313	455	430	2
q14	241	358	256	180	2
q15	531	831	418	409	1
q16	240	374	249	267	1
q17	725	1070	693	655	1
q18	1709	2964	1620	2022	2
q19	63	75	53	54	1
q2	1147	1852	1126	1406	1
q20	1014	1479	992	1271	2
q21	1246	1501	1140	1314	2
q22	45	57	51	50	2
q3	575	345	246	346	2
q4	144	155	115	153	1
q5	386	418	382	504	1
q6	244	323	273	175	1
q7	2166	2580	2342	3013	2
q8	309	340	319	341	2
q9	1582	1750	1427	1819	2

Why this matters to the project

The hardware signature is supposed to capture *the machine*. If a major version change alters plans and runtimes on identical hardware and identical data, then 'production' is a pair — hardware **and** engine version — and a two-scalar hardware signature cannot express it. Any deployment comparing a dev box to a production server on a different PostgreSQL version is measuring both effects at once and attributing all of it to hardware.

Limitations

- One machine, so this isolates the engine cleanly but says nothing about version x hardware interaction.
- Default configuration per version, so some of the difference is changed *defaults* rather than changed planner logic. Separating those would need each version run with the others' settings explicitly pinned.
- `pg_restore` moves data but statistics are rebuilt by `analyze` on each version, so small cost differences may be sampling noise in the statistics rather than planner changes.